

Internal networking is a great way to learn about various networking techniques and methods. Of course if you don't have hell lot of memory (RAM) installed, you can't run too many machines in one go. Also, virtual machines are not going to emulate routers, hubs, switches and firewalls for you; nothing configurable to a decent extent at all! For learning about them, it would be great to get yourself some network emulation software.

If you're a security enthusiast and wanted to try out and learn about how to protect machines from attacks, virtualization is the way to go. You would not be breaking laws (we would not recommend or want our readers to fall into problems) and it is all safe with the data because you're just working with "virtual" systems. With snapshots put to use creatively, you can use virtual machines in quite a lot of useful manners to learn about networking from the operating system scenario. As such, virtual machines provide you an excellent platform for network security auditing of single machines.

Cross platform development

Save web development, almost every type of software development needs some cross platform testing. Even if you're simply coding an application which uses *the .NET* framework and runs on Windows XP, Windows Vista and Windows 7, you might just want to see that it runs fine on all three. In such a scene, you're in need of three operating systems and every single time you need to change something in the program code and test whether the change doesn't break the working in the different operating systems, you need to either reboot the computer (in case all the operating systems are installed on one single machine) or switch between the machines (if you installed the operating systems on three different computers). You either need to have three machines and a lot of space or you need to have a lot of time and patience to keep rebooting your machine! The situation gets more complex if you're developing something more cross platform which works on all supported versions of Windows and Mac as well as Linux. More operating systems and versions you try to support, more the headache you're calling for.

Let's stick only with *the .NET* development scene we discussed and compare that to virtualization – you can have three operating systems running at the same time as virtual machines. Keep your work on one and use the internal networking to make sure that your code is available on all the three virtual machines. Neither are you going to need a lot of time to keep rebooting the machine, nor are you going to be in need of three machines. All you need